

NAME: \_\_\_\_\_

PERIOD: \_\_\_\_\_

## One Plot, Two Reporting Strategies

AP Statistics · Unit 5 · Topic 5.1

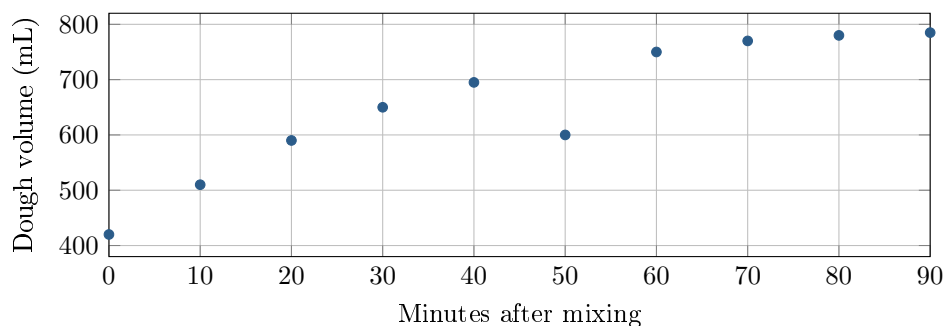
### [STAR] THE PROBLEM

Suppose a test kitchen measures dough volume for ten batches at assigned times after mixing. The scatterplot shows the paired measurements.

**Jules:** Use a straight-line summary because volume generally increases with time. Treat the 50-minute batch as ordinary variation.

**Ravi:** Decide the form from how the rate changes, and report the 50-minute batch separately if it departs from nearby values.

Selected measurements (minutes, mL): (0, 420), (20, 590), (70, 770), (90, 785).



### FIRST TAKE — one sentence before you work the parts

Which reporting strategy seems more defensible, Jules's or Ravi's? Cite one graph feature.

### [BOOK] BUILD A COMPLETE SCATTERPLOT DESCRIPTION (keep on the page)

Put the explanatory variable on the  $x$ -axis and the response on the  $y$ -axis. Describe direction, form, strength, and unusual features in context. Linear form has an approximately constant rate; nonlinear form may curve or change rate. Report an unusual point without letting that one point erase the pattern followed by the rest.

**Work (a)–(c).**

**DOK 1** (a) Name the explanatory and response variables, with units.

**DOK 2** (b) Find the volume increase from 0 to 20 minutes and from 70 to 90 minutes. Is the 50-minute batch above or below the nearby batches?

**DOK 3** (c) Whose reporting strategy is supported? Describe the direction, form, strength, and unusual point in context, using your two increases. Explain how a straight-line summary could mislead someone predicting later growth. Write 3–4 sentences.

**Frame:** I choose \_\_\_\_\_: the pattern is \_\_\_\_\_ in direction, \_\_\_\_\_ in form, and \_\_\_\_\_ in strength.

**Frame:** The two increases, \_\_\_\_\_ and \_\_\_\_\_ mL, show \_\_\_\_\_; a straight line could mislead by \_\_\_\_\_.

**Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.**